

# **CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE**

Team ID : SWTID-2026-1443

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## User Story

The Customer Support Ticket Priority Prediction and Automated Assignment System uses Salesforce and Agentforce to analyze customer support ticket descriptions, automatically determine priority as High, Medium, or Low, and trigger backend automation for appropriate support handling. The solution uses the Support Ticket Intelligence custom object, an Auto-Launched Flow, and an Agentforce subagent named Support Ticket Priority Analysis. High-priority tickets create an urgent handling task and are assigned to a senior support agent.

## 1. INTRODUCTION

### Project Overview

Customer support teams receive a large number of tickets daily, while prioritization and assignment can be manual. This project provides a Salesforce-based solution that analyzes ticket information, classifies urgency, assigns the appropriate support level, and initiates automation for critical cases.

### Purpose

- Automatically classify support tickets as High, Medium, or Low priority.
- Retrieve the latest support ticket associated with a customer Account.
- Reduce manual prioritization and assignment effort.
- Create an urgent handling task for High-priority tickets.
- Use Agentforce to analyze ticket details and trigger backend automation.
- Improve ticket response and resolution efficiency.

## 2. IDEATION PHASE

### Problem Statement

| PS   | User          | I'm trying to                   | But                                                                            | Which makes me feel |
|------|---------------|---------------------------------|--------------------------------------------------------------------------------|---------------------|
| PS-1 | Support Agent | focus on critical tickets first | tickets are not always prioritized automatically; prioritization can be manual | frustrated          |
|      |               |                                 |                                                                                |                     |

|      |                 |                                           |                                                        |             |
|------|-----------------|-------------------------------------------|--------------------------------------------------------|-------------|
| PS-2 | Customer        | get urgent issues resolved quickly        | critical issues may wait behind lower-priority tickets | concerned   |
| PS-3 | Support Manager | assign tickets to the right support level | assignment may require manual decisions                | overwhelmed |
| PS-4 | Support Agent   | identify urgent issues from descriptions  | important keywords can be missed                       | uncertain   |
| PS-5 | Support Manager | reduce SLA-related risk                   | older unresolved tickets may remain unnoticed          | concerned   |

## Empathy Map Canvas

User Persona: Support Agent

- Think & Feel: Wants critical issues identified quickly so urgent customer problems receive attention.
- Hear: Receives customer issue descriptions and support requests.
- See: Ticket status, issue type, description, priority level, assignment, and SLA risk.
- Say & Do: Reviews tickets, handles assigned cases, and focuses on urgent support issues.
- Pain: Manual prioritization, missed urgency signals, and delays in handling critical tickets.
- Gain: Automatic priority classification, clearer assignment, urgent-task creation, and reduced manual effort.

## Brainstorming & Idea Prioritization

- Manual ticket registers
- Spreadsheet-based ticket tracking
- Standalone support portal
- Salesforce CRM with Agentforce

Final Selection: Salesforce CRM selected due to automation through Salesforce Flow, AI-assisted ticket analysis through Agentforce, automatic priority classification, automatic urgent-task creation, and reduced manual support effort.

### 3. REQUIREMENT ANALYSIS PHASE

#### Customer Journey Map

Ticket Creation → Account Identification → Ticket Retrieval → Description Analysis  
→ Priority Classification → Agent Assignment → Urgent Task Creation → Support  
Handling → Resolution

#### Solution Requirements

##### Functional Requirements

| FR No. | Requirement                            |
|--------|----------------------------------------|
| FR-1   | Support ticket creation and management |
| FR-2   | Account and Contact association        |
| FR-3   | Ticket description analysis            |
| FR-4   | Automatic priority classification      |
| FR-5   | High-priority task creation            |
| FR-6   | Support agent assignment               |
| FR-7   | SLA breach risk checking               |
| FR-8   | Agentforce conversational analysis     |

##### Non-Functional Requirements

| NFR             | Description                                                              |
|-----------------|--------------------------------------------------------------------------|
| Usability       | Clear Agentforce interaction and understandable priority messages        |
| Reliability     | Consistent priority classification based on configured conditions        |
| Performance     | Quick retrieval and analysis of the latest support ticket                |
| Security        | Controlled Salesforce access to customer and ticket information          |
| Maintainability | Flow and Agentforce configuration can be updated                         |
| Scalability     | Supports increasing support-ticket volumes through Salesforce automation |

#### Data Flow Diagram

Customer / Support Request → Account → Latest Support Ticket → Ticket Description  
→ Analyze Description → Priority Level → High/Medium/Low Decision → Agent  
Assignment → High-Priority Task (if applicable) → Final Response

Agentforce: User provides Account Name → Agentforce Support Ticket Priority Analysis → Auto-Launched Flow → Ticket Analysis → Priority and Assignment Output → Response to User.

### Technology Stack

| Layer               | Technology                                                                  |
|---------------------|-----------------------------------------------------------------------------|
| Platform            | Salesforce                                                                  |
| Database            | Support_Ticket_Intelligence__c with Account, Contact, User and Task records |
| Logic               | Auto-Launched Flow                                                          |
| AI                  | Agentforce                                                                  |
| Agent Configuration | Support Ticket Priority Analysis subagent                                   |
| Automation          | Flow Decision, Assignment and Create Records                                |
| Output              | Agentforce conversation response                                            |

## 4. PROJECT DESIGN PHASE

### Problem–Solution Fit

| Problem                        | Solution                                    | Benefit                               |
|--------------------------------|---------------------------------------------|---------------------------------------|
| Manual ticket prioritization   | Flow analyzes description and sets priority | Faster identification of urgent cases |
| Urgent issues may be missed    | High-priority keyword decision              | Critical cases receive attention      |
| Manual task creation           | Flow creates Task for High priority         | Reduced manual effort                 |
| Manual support assignment      | Flow sets assigned support level            | More consistent handling              |
| SLA risk can be overlooked     | Optional SLA Risk decision                  | Better risk awareness                 |
| Need for conversational access | Agentforce subagent AI-assisted interaction | Conversational support handling       |

### Proposed Solution

| Parameter | Description                                                                                             |
|-----------|---------------------------------------------------------------------------------------------------------|
| Problem   | Manual ticket prioritization and assignment causing delays in handling critical customer support issues |
| Solution  | Salesforce-based Customer Support Ticket Priority Prediction and Automated Assignment System            |

|                |                                                                                                                                                               |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Innovation     | Agentforce AI + Auto-Launched Flow for analyzing ticket descriptions, classifying priority, assigning support level, and triggering urgent-task automation    |
| Impact         | Faster identification of critical tickets, reduced manual effort, improved response time, and better support-team productivity                                |
| Business Model | Customer support operations involving customers, support agents, support managers, and AI-assisted ticket handling                                            |
| Scalability    | Supports increasing support-ticket volumes through Salesforce automation, with scope for additional assignment rules, SLA monitoring, and priority indicators |

### Solution Architecture

Users / Support Team → Agentforce → Support Ticket Priority Analysis → Auto-Launched Flow → Account & Ticket Data → Description Analysis → Priority Decision → Task / Assignment → Final Message → Agentforce Response

## 5. PROJECT PLANNING & SCHEDULING

### Project Planning

Agile methodology is used with Epics, User Stories, Story Points, and sprint-based execution. Velocity calculation is used for estimation.

### Product Backlog

| Sprint / Epic            | User Story                                                                                  | Points | Priority |
|--------------------------|---------------------------------------------------------------------------------------------|--------|----------|
| Sprint-1 Developer Setup | USN-1: Configure Salesforce environment for Support Ticket Intelligence automation solution | 3      | High     |
| Sprint-2 Data Modeling   | USN-2: Create Support Ticket Intelligence object with required fields and relationships     | 5      | High     |
| Sprint-2 Data            | USN-3: Associate                                                                            | 3      | High     |

|                                    |                                                                                                 |   |        |
|------------------------------------|-------------------------------------------------------------------------------------------------|---|--------|
| Modeling                           | support tickets with Accounts and Contacts                                                      |   |        |
| Sprint-3 Automation                | USN-4: Create Auto-Launched Flow to retrieve latest ticket and analyze description              | 5 | High   |
| Sprint-3 Automation                | USN-5: Classify tickets as High, Medium, or Low using urgency conditions                        | 5 | High   |
| Sprint-4 Agentforce & Automation   | USN-6: Analyze support ticket details and trigger backend Flow using Agentforce                 | 8 | High   |
| Sprint-4 Task & Assignment         | USN-7: Create urgent task and assign appropriate support level for High priority                | 5 | High   |
| Sprint-5 SLA Management            | USN-8: Check SLA breach risk for older unresolved tickets                                       | 5 | Medium |
| Sprint-6 Agentforce & Conversation | USN-9: Provide Account Name and receive ticket ID, priority, assigned agent, and action message | 5 | Medium |

## 6. PROJECT DEVELOPMENT PHASE

### Milestone 1: Salesforce Account

Create and activate a Salesforce Developer Org using the required developer signup and activation process.

## Milestone 2: Data Management – Custom Object

Object Name: Support Ticket Intelligence

API Name: Support\_Ticket\_Intelligence\_\_c

| Field Label           | API Name           | Data Type        | Description                 |
|-----------------------|--------------------|------------------|-----------------------------|
| Ticket Number         | Ticket_Number__c   | Auto Number      | TKT-{0000}                  |
| Customer              | Customer__c        | Lookup (Account) | Related Account             |
| Contact               | Contact__c         | Lookup (Contact) | Customer contact            |
| Issue Type            | Issue_Type__c      | Picklist         | Technical, Billing, General |
| Description           | Description__c     | Long Text Area   | Issue details               |
| Priority Level        | Priority_Level__c  | Picklist         | Low, Medium, High           |
| Status                | Status__c          | Picklist         | New, In Progress, Resolved  |
| Created Date          | Created_Date__c    | Date             | Ticket date                 |
| Assigned To           | Assigned_To__c     | Lookup (User)    | Support agent               |
| SLA Breach Risk       | SLA_Breach_Risk__c | Checkbox         | Risk flag                   |
| Resolution Time (hrs) | Resolution_Time__c | Number           | Time taken                  |

### Auto-Launched Flow

Flow Type: Auto-Launched Flow (No Trigger or called from Agentforce).

| Variable       | API Name         | Type | I/O    |
|----------------|------------------|------|--------|
| Account Name   | varAccountName   | Text | Input  |
| Account Id     | varAccountId     | Text | Output |
| Ticket Id      | varTicketId      | Text | Output |
| Priority Level | varPriorityLevel | Text | Output |
| Assigned Agent | varAssignedTo    | Text | Output |
| Action Message | varActionMessage | Text | Output |

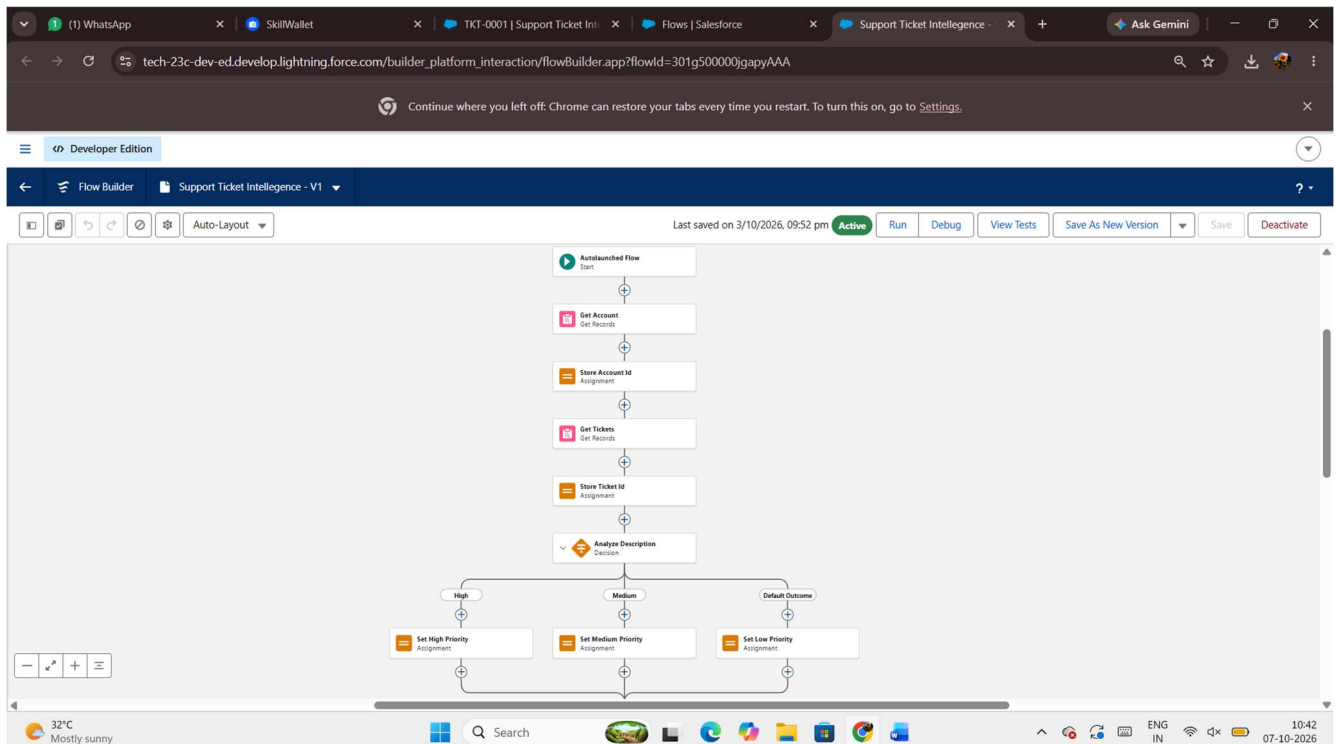
### Flow Implementation

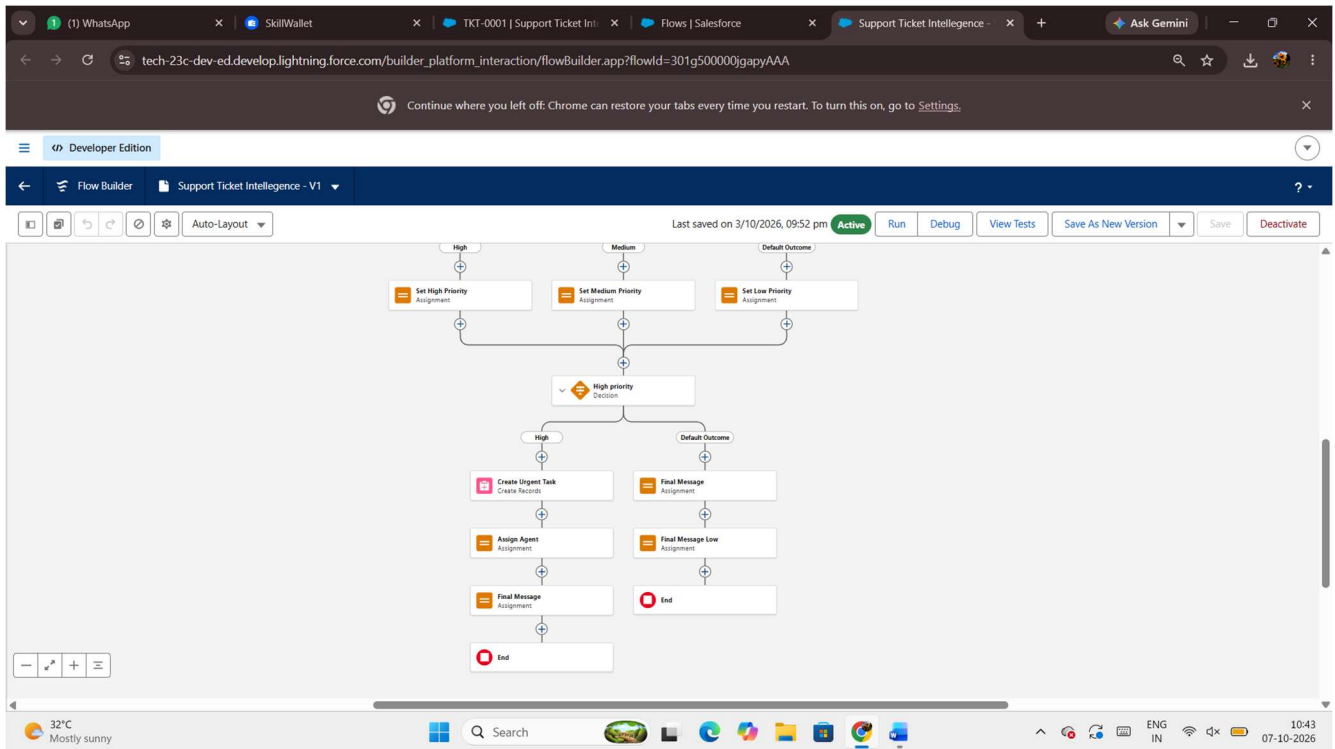
| Step | Element          | Configuration                                                                              |
|------|------------------|--------------------------------------------------------------------------------------------|
| 1    | Get Account      | Account; Name = varAccountName; CreatedDate DESC; first record                             |
| 2    | Store Account Id | varAccountId = Get_Account.Id                                                              |
| 3    | Get Ticket       | Support_Ticket_Intelligence__c; Customer__c = varAccountId; CreatedDate DESC; first record |
| 4    | Store Ticket Id  | varTicketId = Get_Ticket.Id                                                                |

|    |                     |                                                                                     |
|----|---------------------|-------------------------------------------------------------------------------------|
| 5  | Analyze Description | High: urgent/not working/failure; Medium: issue/slow/delay; default: Low            |
| 6  | Set Priority        | Set varPriorityLevel to High, Medium, or Low                                        |
| 7  | High Priority Check | varPriorityLevel = High                                                             |
| 8  | Create Task         | Subject Urgent Ticket Handling; WhatId Ticket Id; Priority High; Status Not Started |
| 9  | Assign Agent        | varAssignedTo = Senior Support Agent                                                |
| 10 | Optional SLA Check  | Created Date older than 2 days → optional internal flag/message                     |
| 11 | Final Message       | High: urgent handling; Medium: handled shortly; Low: queued                         |
| 12 | Save Flow           | Support Ticket Intellegence                                                         |

## Final Flow Structure

Start → Get Account → Assignment (Account Id) → Get Ticket → Assignment (Ticket Id) → Decision (Analyze Description) → Assignment (Priority Level) → Decision (High Priority) → Create Task → Assignment (Assigned Agent) → Optional SLA Decision → Assignment (Final Message) → End





Developer Edition

Search...

Agentforce Studio | TKT-0001 | Support Ticket Intelligence

TKT-0001

Related | Details

| Field          | Value                                    |
|----------------|------------------------------------------|
| Ticket Number  | TKT-0001                                 |
| Owner          | Suriya prabha M                          |
| Customer       | <a href="#">Test Customer</a>            |
| Contact        | <a href="#">Test customer</a>            |
| Issue Type     | Technical                                |
| Description    | system is not working and this is urgent |
| Priority Level |                                          |
| Status         | New                                      |
| Created Date   | 05/10/2026                               |
| Assigned To    |                                          |

Agentforce

Let's chat!

Hi! I'm Einstein, an AI assistant. I can do things like search for information, summarize records, and draft and revise emails. What can I help you with?

"Analyze this ticket for priority."

Describe your task or ask a question...

Agentforce is AI and can make mistakes.

31°C Mostly sunny

Search

ENG IN 1040 07-10-2026

## Final Flow Outcome

- Takes Account Name as input.
- Fetches latest support ticket.
- Analyzes description using keywords.
- Sets priority automatically.
- Creates task for urgent cases.
- Returns output to Agentforce.

## 7. AGENTFORCE CONFIGURATION

### Agentforce Subagent

| Configuration              | Value                                                                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                       | Support Ticket Priority Analysis                                                                                                                       |
| API Name                   | Support Ticket Priority Analysis                                                                                                                       |
| Classification Description | Analyzes customer support tickets to determine priority based on issue description and urgency.                                                        |
| Scope                      | Only analyze ticket descriptions, determine High/Medium/Low, and trigger backend automation for task assignment; unrelated requests are outside scope. |

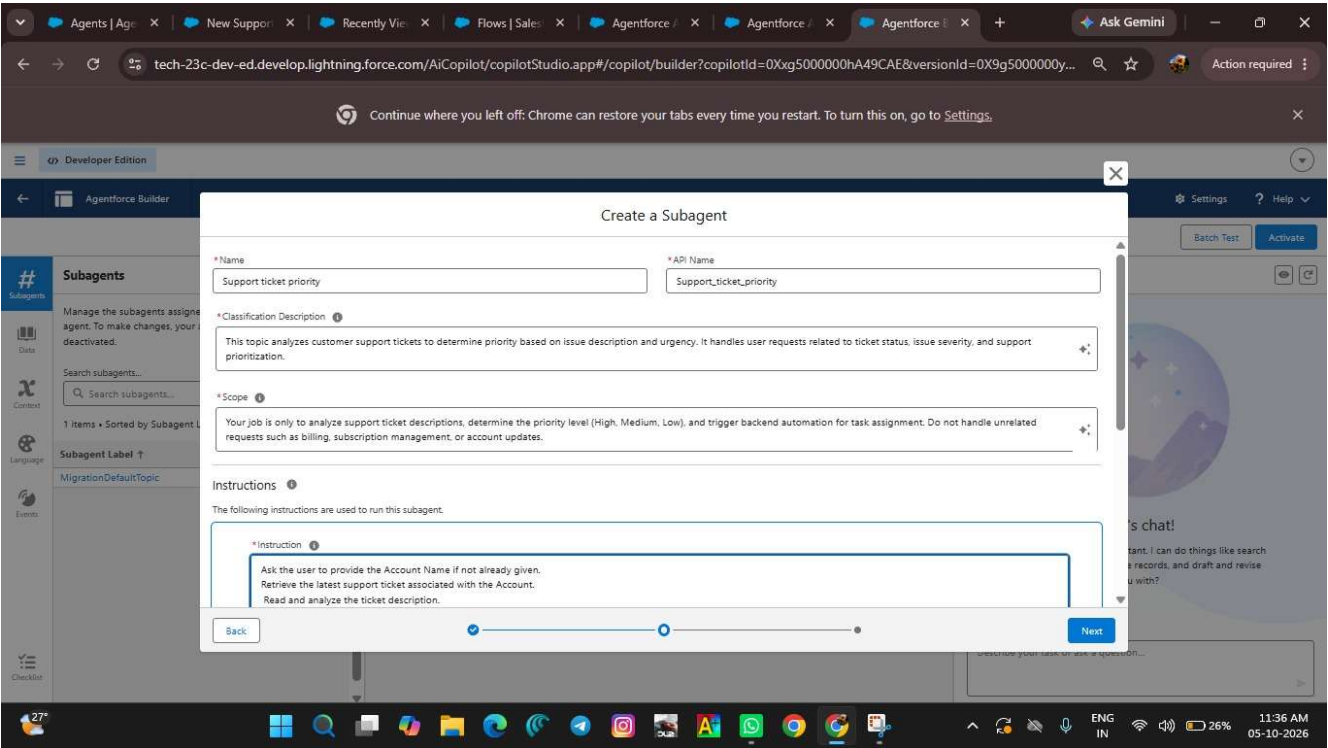
### Instructions

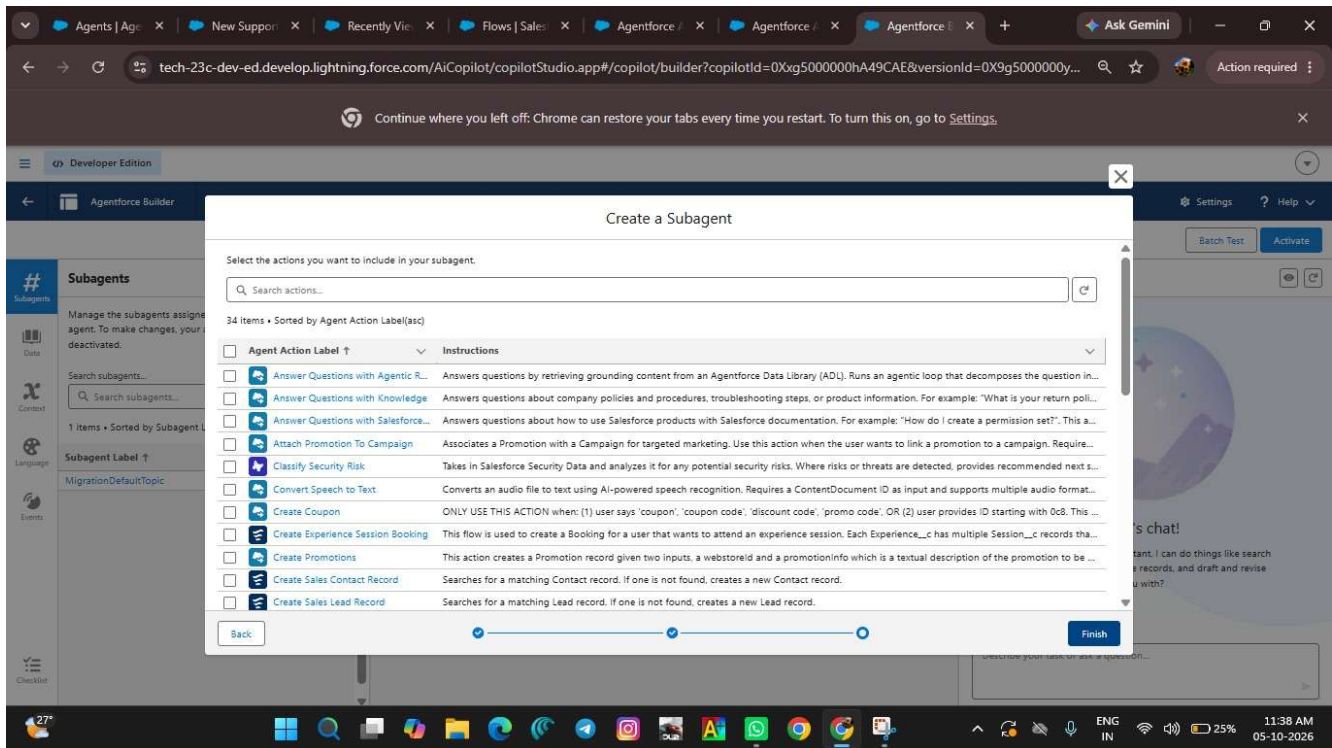
- Ask for Account Name if not given.
- Retrieve the latest support ticket associated with the Account.
- Read and analyze the ticket description.
- High: urgent, not working, failure.
- Medium: issue, slow, delay.
- Low: none of the above.
- Set priority accordingly.
- For High, trigger the Auto-Launched Flow and ensure a task is created.
- Assign appropriate support level, e.g. senior agent for High.
- Return clear High/Medium/Low action messages.
- Do not update records directly or send emails; use Flow for actions.

## Agent Action

Description: Analyzes customer support tickets based on Account Name, determines priority using issue description, and triggers automation for high-priority cases.

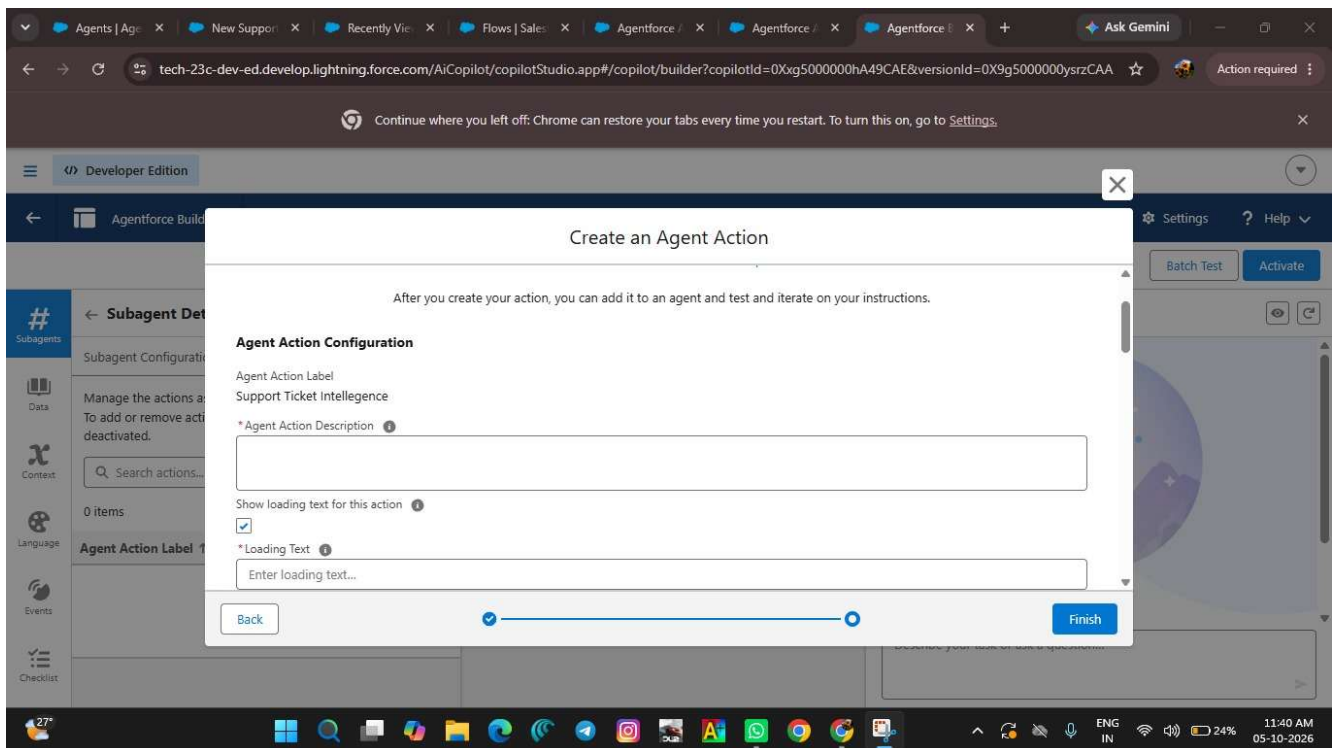
| Type   | API Name         | Data Type           | Setting                                  |
|--------|------------------|---------------------|------------------------------------------|
| Input  | varAccountName   | lightning__textType | Require Input;<br>Collect Data From User |
| Output | varAccountId     | lightning__textType | Show in<br>Conversation                  |
| Output | varActionMessage | lightning__textType | Show in<br>Conversation                  |
| Output | varAssignedTo    | lightning__textType | Show in<br>Conversation                  |
| Output | varPriorityLevel | lightning__textType | Show in<br>Conversation                  |
| Output | varTicketId      | lightning__textType | Show in<br>Conversation                  |





## 8. FUNCTIONAL & PERFORMANCE TESTING

| Test Case | Condition                            | Expected Result                                    |
|-----------|--------------------------------------|----------------------------------------------------|
| TC-01     | Description contains urgent          | High; urgent task; senior support handling         |
| TC-02     | Description contains not working     | High; urgent task                                  |
| TC-03     | Description contains failure         | High; urgent task                                  |
| TC-04     | Description contains issue           | Medium; handled shortly                            |
| TC-05     | Description contains slow            | Medium; handled shortly                            |
| TC-06     | Description contains delay           | Medium; handled shortly                            |
| TC-07     | No configured keyword                | Low; queued                                        |
| TC-08     | Older unresolved ticket, SLA enabled | SLA risk flag/message                              |
| TC-09     | Agentforce receives Account Name     | Latest ticket and priority/action outputs returned |



## 9. ADVANTAGES & DISADVANTAGES

### Advantages

- Automatic Ticket Prioritization
- Faster identification of critical issues
- Reduced Manual Effort
- Intelligent Ticket Handling
- Automatic Urgent Task Creation
- Improved Support Assignment
- Conversational Support
- Better Response Time
- SLA Risk Awareness
- Scalable Support Operations
- Improved Team Productivity

### Disadvantages

- Keyword-Based Priority Logic
- Limited Understanding of Context
- Dependence on Accurate Ticket Data
- Flow Configuration Dependency
- Agentforce Configuration Dependency

- Limited Scope of Agentforce
- SLA Monitoring is Limited
- Assignment Logic May Need Expansion
- Maintenance Required
- Limited Analytics in Current Scope

## 10. CONCLUSION

The Customer Support Ticket Priority Prediction and Automated Assignment System demonstrates how Salesforce Flow and Agentforce can work together to reduce manual support-ticket handling. The solution retrieves the latest ticket for an Account, analyzes its description, classifies priority, creates an urgent task for High-priority cases, assigns a support level, and returns a clear conversational response. This supports faster ticket handling, reduced manual effort, intelligent prioritization, and better team productivity.

## 11. FUTURE SCOPE

- Add more priority indicators and business rules.
- Extend Agentforce outputs with additional ticket information.
- Add more sophisticated SLA monitoring and escalation.
- Support additional support teams or assignment rules.
- Introduce analytics for priority trends and resolution performance.
- Add more variables and configure them in the Agentforce action for richer outputs.

## 12. APPENDIX

- A. Custom Object: Support Ticket Intelligence
- B. Key Salesforce Records: Account, Contact, User, Task
- C. Flow: Support\_Ticket\_Intelligence – Auto-Launched Flow
- D. Agentforce Subagent: Support Ticket Priority Analysis
- E. Agentforce Action: Flow-based Support Ticket Intelligence action